
MA 6040 - Fuzzy Logic Connectives: Theory and Applications
Assignment 4 - Fuzzy Relations - 29, September 2026

1. Let P, Q, R be fuzzy relations on $X \times Y, Y \times Z, X \times Z$, respectively and let $\overset{T}{\circ}$ denote the max- T -composition, where T is a t-norm. Show the following:

(a) $(P \overset{T}{\circ} Q) \overset{T}{\circ} R = P \overset{T}{\circ} (Q \overset{T}{\circ} R).$

(b) $P \overset{T}{\circ} \left(\bigcup_{i \in I} Q_i \right) = \bigcup_{i \in I} (P \overset{T}{\circ} Q_i).$

(c) $P \overset{T}{\circ} \left(\bigcap_{i \in I} Q_i \right) \subseteq \bigcap_{i \in I} (P \overset{T}{\circ} Q_i).$

(d) $(P \overset{T}{\circ} Q)^{-1} = Q^{-1} \overset{T}{\circ} P^{-1}$, where A^{-1} refers to the transpose of A .

2. Let R be a reflexive fuzzy relation on $X \times X$ and let $|X| = n$ s.t. $n \in \mathbb{N}$ and $n \geq 2$. Show that the transitive closure of R w.r.t. the max- T -composition is, in fact, $R^{[n-1]}$, i.e., R composed with itself $n - 1$ times.
3. Check if the following fuzzy relations are transitive or not. If they are not, find their transitive closure:

$$A = \begin{bmatrix} 0.1 & 1.0 & 0.5 & 0.9 \\ 0.3 & 0.2 & 0.8 & 0.8 \\ 0.5 & 1.0 & 0.1 & 1.0 \\ 1.0 & 1.0 & 0.4 & 0.7 \end{bmatrix} \quad B = \begin{bmatrix} 1.0 & 0.8 & 0.55 & 0.35 \\ 0.8 & 1.0 & 0.55 & 0.4 \\ 0.55 & 0.55 & 1.0 & 0.55 \\ 0.35 & 0.4 & 0.55 & 1.0 \end{bmatrix}$$

4. Consider the relation R on $X = [0, 1]^2$ defined as:

$$R(x, y) = e^{-||x-y||_p}, \quad x, y \in [0, 1]^2$$

where $||x - y||_p$ is the p -norm. Determine t-norm(s) T w.r.t. to which it satisfies T -transitivity for $p = 1, 2$.

5. Consider the following relations R_i on a suitable discrete universe of discourse X_i :

$$R_1 = \begin{bmatrix} 1 & 0.8 & 0 & 0 & 0 & 0 & 0 \\ 0.8 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0.8 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0.8 & 0.7 & 0.5 \\ 0 & 0 & 0.8 & 0.8 & 1 & 0.7 & 0.5 \\ 0 & 0 & 0 & 0.7 & 0.7 & 1 & 0.4 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0.4 & 1 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} 1 & 0.6 & 0.6 & 0.5 & 0.9 & 0.8 & 0.6 & 0.7 & 0.7 \\ 0.6 & 1 & 0.6 & 0.5 & 0.6 & 0.6 & 0.6 & 0.6 & 0.6 \\ 0.6 & 0.6 & 1 & 0.5 & 0.6 & 0.6 & 0.6 & 0.6 & 0.6 \\ 0.5 & 0.5 & 0.5 & 1 & 0.5 & 0.5 & 0.5 & 0.5 & 0.5 \\ 0.9 & 0.6 & 0.6 & 0.5 & 1 & 0.8 & 0.6 & 0.7 & 0.7 \\ 0.8 & 0.6 & 0.6 & 0.5 & 0.8 & 1 & 0.6 & 0.7 & 0.7 \\ 0.6 & 0.6 & 0.6 & 0.5 & 0.6 & 0.6 & 1 & 0.6 & 0.6 \\ 0.7 & 0.6 & 0.6 & 0.5 & 0.7 & 0.7 & 0.6 & 1 & 0.7 \\ 0.7 & 0.6 & 0.6 & 0.5 & 0.7 & 0.7 & 0.6 & 0.7 & 1 \end{bmatrix}$$

Check if R_i is a similarity/ T -equivalence relation w.r.t. T_M . If yes, find partition of X w.r.t. $\alpha = 0.5$. If no, find compatibility classes of X w.r.t. $\alpha = 0.5$.

6. Implement the following algorithms in Matlab:

- (a) Finding the $\max - T$ -composition of given two fuzzy relations, where T can be chosen by the user from any one of the available choices.
- (b) Finding the transitive closure of a binary fuzzy relation R .
- (c) Given a fuzzy relation R , make it reflexive and symmetric to obtain a compatibility relation. Obtain its compatibility classes.
- (d) Given a compatibility relation, make it a similarity relation and find its similarity classes.
- (e) Verify your implementation with the above fuzzy relations from the previous problems.

7. Can you find an A such that the following fuzzy relational equation is valid?

$$A \circ \begin{bmatrix} 0.1 & 0 & 0.8 \\ 0.4 & 0.6 & 0.5 \\ 0.9 & 0.8 & 0 \end{bmatrix} = [0.8 \quad 0.8 \quad 0.3] \text{ .}$$